

Examiner
only

- (i) **Complete the Punnett square** below to show the possible genotypes of the future children of person **1** and person **2**. Use the letters **N** to represent the dominant allele and **n** to represent the allele that causes cystic fibrosis. [2]

- (ii) Use the Punnett square to predict the probability of person **1** and person **2** having another child who has cystic fibrosis. [1]

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- (d) One treatment for cystic fibrosis is to introduce dominant alleles (**N**) into the cells lining the lungs.

- (i) State the name of this type of treatment. [1]

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- (ii) State the method that is used to deliver the dominant alleles to the lungs. [1]

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- (iii) State **one** problem with the use of this method. [1]

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2. Students used a quadrat to investigate the abundance of dandelions (*Taraxacum officinale*) on the school rugby pitch.



The method they used was:

1. Use a 1 m² quadrat.
2. Use a random number generator to place the quadrat on the rugby pitch.
3. Count the number of dandelions in the quadrat.
4. Repeat steps 2 and 3 another 5 times.
5. Calculate a mean.
6. Calculate the number of dandelions on the school rugby pitch.

The results they obtained are given in **Table 2.1**.

Table 2.1

Quadrat	Number of dandelions
1	3
2	5
3	2
4	7
5	15
6	6
Mean	



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- (a) (i) **Draw a circle** around the anomalous result in **Table 2.1**. [1]
- (ii) Calculate the mean number of dandelions per quadrat and **write your answer in Table 2.1. Do not use the anomalous result in your calculation.** [2]

- (b) The teacher said that he did not have confidence in the mean. Suggest why. [1]
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- (c) The total area of the school rugby pitch is 7 350 m². Use your answer to part (a)(ii) to calculate the total number of dandelions on the school rugby pitch. [2]

Total number of dandelions on the school rugby pitch =

- (d) State the name of the method the students should use to investigate the distribution of the dandelions across the pitch. [1]
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